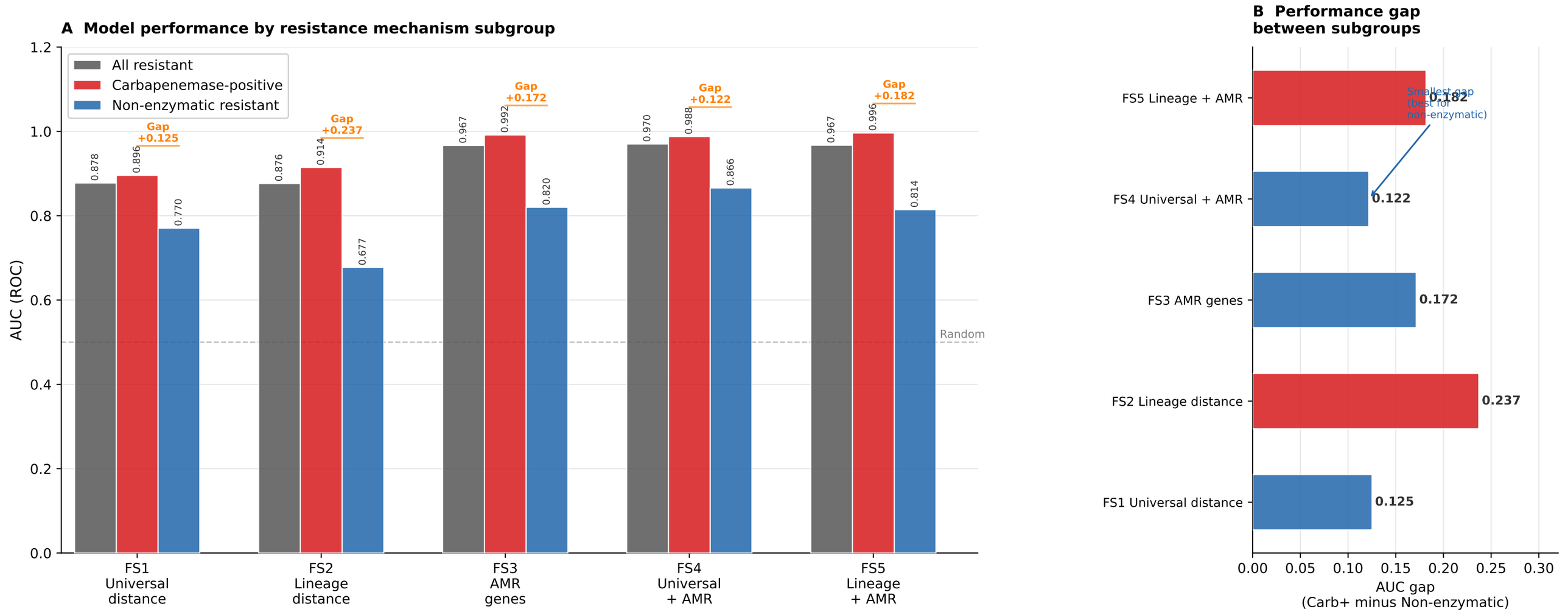
Subgroup Analysis: Carbapenemase-Positive vs Non-Enzymatic Resistant Isolates
XGBoost model evaluated on held-out test set



Carbapenemase-positive: isolates with blaOXA-23, blaOXA-58, blaOXA-24, blaNDM-1, or blaVIM detected (n=262 resistant, test set n≈52)
Non-enzymatic resistant: resistant isolates with no detected acquired carbapenemase (n=51 total, test set n≈10)
All models trained on full training set (resistant vs susceptible); evaluated on test set stratified by resistance mechanism